

Audio Recognition Research Methodology

This document outlines the research methodology used in our audio recognition system.

1. Data Collection and Preprocessing

- Multilingual speech corpus across 6+ languages
- Environmental sound recordings from diverse settings
- Acoustic event classification datasets
- Audio mixing techniques for realistic scenarios

2. Model Architecture

- Transformer-based encoder-decoder architecture
- Multi-head attention mechanisms for audio feature extraction
- Joint training on speech and non-speech audio

3. Training Approach

- Transfer learning from pre-trained audio models
- Contrastive learning for improved feature discrimination
- Fine-tuning on domain-specific tasks

4. Evaluation Metrics

- Word Error Rate (WER) for speech recognition
- F1 Score for event detection
- Mean Average Precision (mAP) for audio classification